

further inputs by just prompting '>'. Check below for how to create the 3x3 magic square matrix, followed by various other interesting operations:

```
$ octave -qf
octave:1> M = [ # 3x3 magic square
> 8 1 6
> 3 5 7
> 4 9 2
> ]
M =

    8    1    6
    3    5    7
    4    9    2

octave:2> B = rand(3, 4); # 3x4 matrix w/ randoms in [0 1]
octave:3> B
B =

    0.068885    0.885998    0.542059    0.797678
    0.652617    0.904360    0.036035    0.737404
    0.043852    0.579838    0.709194    0.053118

octave:4> B' # Transpose of B
ans =

    0.068885    0.652617    0.043852
    0.885998    0.904360    0.579838
    0.542059    0.036035    0.709194
    0.797678    0.737404    0.053118

octave:5> A = inv(M) # Inverse of M
A =

    0.147222   -0.144444    0.063889
   -0.061111    0.022222    0.105556
   -0.019444    0.188889   -0.102778

octave:6> M * A # Should be identity, at least approx.
ans =

    1.00000    0.00000   -0.00000
   -0.00000    1.00000    0.00000
    0.00000    0.00000    1.00000

octave:7> function rv = psine(x) # Our phase shifted sine
>
> rv = sin(x + pi / 6);
>
> endfunction
octave:8> x = linspace(0, 2*pi, 400); # 400 pts from 0 to 2*pi
octave:9> plot(x, psine(x)) # Our function's plot
```

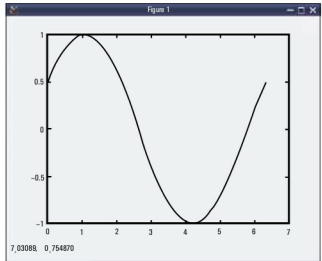


Figure 1: Plot of our 30° phase shifted sine

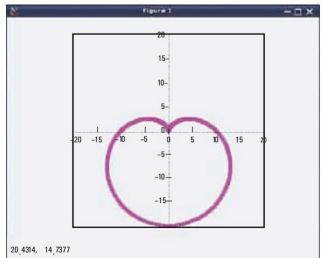


Figure 2: The heart

```
octave:10> polar(x, 10 * (1 - sin(x)), 'm') # bonus Heart
octave:11> quit
```

Figure 1 shows the plot window that pops up in response to command #9 – plot and Figure 2 is the bonus magenta heart of stars (*) from polar coordinates draw command #10 – polar.

What next?

With a ready-to-go level of introduction to *octave*, we are all set to explore it the fun way. What fun, you may well ask. That's left to your imagination. And as we move on, you could take up one or more fun challenge(s) and try to solve them using *octave*. **END** 

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